A Comparative Analysis of Hybrid Quantum Neural Networks in Binary Credit Defaulting Tasks

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***Abstract*—Quantum machine learning has been shown to offer exponential computational boosts compared to current classical models through the exploitation of quantum mechanical properties such as entanglement and superposition. The finance sub field of fin-tech, where financial services are constructed around a customer's demands, is believed to benefit with the use of hybrid quantum-classical machine learning algorithms. Classical algorithms struggle to create accurate, complex, and prompt financial analytical models resulting in over 25% of small and medium sized financial institutions losing customers each year. This study explores whether a hybrid quantum neural network (QCNN) design provides a quantum advantage in either accuracy or time in the fin tech task of credit defaulting. The hybrid quantum neural network model uses a variational quantum circuit built on a continuous variable architecture. The data flow of the network consists of a classical network, quantum data encoding (which uses quantum squeezers, interferometers, displacement, and Kerr gates), QCNN, and measurement. Although the quantum model took a longer time to train, it reported a higher prediction accuracy of 97.63% compared to 91.20% for the classical network. This work demonstrates the future implications of quantum integration in the financial field and the quantum advantage it offers on classification tasks.**

***Index Terms*—credit scoring, feature selection, computational finance, variational circuit, quantum machine learning**

I. Introduction

Since the 1980s, the field of quantum computing has attracted the attention of researchers in its ability to solve complex problems, with the exploitation of quantum mechanical properties [14]. Scientists, such as Fenyman were among the first to suggest that quantum computing algorithms may outperform their classical counterpart algorithms [14]. Quantum computing, in comparison to classical computing, relies on a fundamentally different way of processing and storing information in terms of the laws of quantum mechanics including superposition, entanglement, and interference [9]. Using these properties, quantum algorithms have resulted in exponential computation boosts and enable the simultaneous processing of more inputs in fewer operations, in comparison to classical algorithms [16]. One of the first fields to benefit from quantum algorithms is believed to be finance as many of its cases can be suited for near-term quantum computers [1]. Many financial problems can be formulated in terms of non-convex optimization or NP complete problems with which the computational cost associated with classical techniques is too high and machine learning algorithms, such as neural networks, are prone to overfitting [2]. Today's customers demand personalized products and services that swiftly adapt to their financial needs and behaviors [19]. However, classical algorithms struggle to create analytical models that cannot only quickly and accurately analyze substantial data, but also examine complex non-linear interactions [8]. Resultantly, over 25% of small and medium sized financial institutions lose their customers [19]. Hybrid quantum neural network systems have the potential to offer significant advantages in addressing financial problems, particularly in the context of credit defaulting. This powerful quantum machine learning approach can be effectively applied in large data processing, risk management, and classification, especially when dealing with crucial information related to a group of interest [9].

II. Literature Research

Quantum computing is an emerging technological field that is the study of various information processing tasks that can be performed using quantum mechanical systems and properties [14]. Attention has shifted to this field as computer science and its conventional approaches in the development of computer technology are beginning to oppose fundamental classical physics laws [14]. Researchers believe that quantum computers offer a significant speed advantage over classical computers that cannot be overcome by classical computers [14]. Additionally, with a reduction in time and memory required for computational tasks, quantum advantages could lead to unparalleled scalability and accuracy in computations [9]. However, the quantum computing era is currently defined as Noisy Intermediate Scale Quantum (NISQ), in which complications exist such as noise and decoherence, limiting the capability and usage of near-term quantum computers [18]. Despite these limitations, quantum machine learning (QML), a subfield of quantum computing, has been shown to provide exponential speed-ups over classical machine learning algorithms in learning tasks by utilizing its ability to manipulate high dimensional vectors [12]. This ability is crucial

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as the field of machine learning relies on vectors and tensor products to represent large amounts of data [12]. Operations that manipulate vectors such as dot products, overlaps, and norms in N-dimensional vector spaces require a time of O(N) in classical machine learning algorithms compared to O(log N) in QML algorithms [12]. Moreover, post-processing operations such as Fourier transformations and matrix inversions require a time of O(poly(N)) compared to O(poly(log(N)), for a classical and quantum machine learning algorithm, respectively [12]. One of the fields that is believed to benefit from the quantum advantage in QML is finance, as its applications have started to become more extensive [3].

Many finance problems, such as portfolio optimization and option pricing, are fundamentally non-convex optimization problems [6]. However, classical machine learning algorithms are prone to overfitting and are typically classified as NP complete problems [2]. As a result, current classical algorithmic approaches often become computationally impractical and expensive [15]. Quantum approaches, on the other hand, find a natural use as many of the problems are equivalent to finding the ground state of a Hamiltonian [7]. Instead, quantum algorithms face challenges in implementing non-linearities in classical algorithms, like neural networks, as quantum algorithms tend to be linear and unitary [21]. Moreover, quantum machine learning algorithms tend to presume the abundant presence of a quantum random access memory (qRAM) to encode classical information and store it [3]. Yet, quantum hardware is at a stage where efficient qRAM fails to exist and is still in development [15].

One viable solution is the incorporation of quantum devices in classical neural network models: Hybrid Quantum Classical Neural Networks (QCNN). Classical neural networks in deep learning inter web multiple layers of nonlinear and linear functions leading to computational complexity [11]. Consequently, these networks suffer from large training times for substantial data sets as the number of training parameters increases [11]. QCNN addresses these problems by employing classical layers to capture large-scale features while preserving nonlinearity and the locality of a convolutional layer in a deep learning model [11, 20]. Additionally, it utilizes quantum layers to represent linearity and harness the capabilities of NISQ technology [11].

Variational Quantum Algorithms (VQA) have emerged as a prominent analogue to the quantum layers in a deep learning model and are the most promising avenue to achieving a quantum advantage [4]. VQAs have been proposed and explored in practically every sector with finance being a plausible application. For instance, Schetakis et al. [20] explore the usage of a full hybrid quantum neural network model consisting of Variational Circuit Classifiers (VC) and data reuploading classifiers (DRC) on credit scoring for Singaporean small and medium businesses. The initial layers of the model consist of classical rectified linear unit (ReLU) layers [20] The experiment's VC used a data embedding layer followed with parameterized gates and entangling layers. At the same time, the DRC replicated the VC into more blocks in the model to provide the ability to capture nonlinear patterns with robustness to noise [20]. As a result, the full hybrid algorithm outputted an ROC of 0.75 compared to an ROC output of 0.74 for the classical algorithm with 350 epochs run for each [20]. It was also observed that the classical algorithm would be able to achieve the same ROC output with 5,000 epochs, a significantly higher run value [20].

Arthur and Date [2] used a different VC on various binary classification datasets (such as the Iris benchmark dataset). The feature map of the VC was used to encode the features by restricting the values of the feature set to [0, π] and using the resulting value to encode it into a corresponding amplitude of a qubit [2]. This encoding ensures the preservation of values in the features through a combination of rotations and measurements [2]. The ansatz of the VC used Qiskit's RealAmplitude ansatz with one repetition and full entanglement using RY rotation and CX entanglement gates. [2]. The overall VC's cost is minimized using gradient descent calculated with the parameter shift rule [2]. In conjunction with the VC, the hybrid neural network model had a single hidden layer with one output unit which helped to reduce computational complexity [2]. Although the hybrid QCNN had longer training times, it tended to always outperform the individual VC in accuracy in addition to cost by eight to 11 percent and 20 to 40 percent, respectively [2].

Killoran et al. [10] and Choe [5] both explored fully connected layer structure from classical layers to a continuous variable (CV) quantum architecture. Killoran et al. [10] developed fully connected quantum layers in which their gates consist of a sequence of universal gate sets of N-port linear optical interferometers containing beamsplitter and rotation gates as well as non-Gaussian gates. In addition to a classical network layer, which adds non-linearity, the quantum circuit remains unitary. Choe [5] built a similar hybrid architecture for image classification in which classical feed forward layers are used to preprocess the data and then reduced to smaller vectors through the exponential linear unit (ELU) activation function followed by a specific sequence of Kerr, displacement, interferometer, squeezer, and interferometer gates. Both models performed highly with Killoran et al. [10] circuit's outputs resembling very closely to noiseless ground truth curves for the task of curve fitting along with Choe [5] achieving an accuracy of 99.90% on the MNIST dataset with an eight class classifier [5, 10].

Variational circuits have been shown to offer a promising advantage over classical algorithms in various applications, including finance. More importantly, variational circuits are a powerful toolbox that hold the capability of smoothly integrating into pre-existing classical neural network models and overtaking the linear segments of the model. Thus, allowing the classical portion of the model to perform nonlinear transformations and the quantum portion of the same model to excel in mathematical linear transformations (i.e. vector and matrix transformations). With the exploitation of the computing power for both classical and quantum approaches, the hybridization offers the ability to process large, complex

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datasets in an accurate and timely manner - an ability that the financial field lacks. This study aims to prove the integration of quantum technology, specifically variational circuits, in classical neural networks models and bridge the gap of whether quantum integration offers a quantum advantage in processing financial tasks.

III. Experimental Design

A. Dataset Collection

The dataset used in this study was prepared by Fransiscus Pratikto of Universitas Katolik Parahyangan Jurusan Teknik Industri and published in March of 2023 [17]. The dataset contains over 17,000 cases of credit card holder data. The data was stored in a Microsoft Excel file.

B. Dataset Features and Analysis

The data set has 20 predictor features and one binary target feature. These predictor features include trivial monetary data such as the number of cards a client has, their sex, age, education level, history of past payments, credit amount, etc. Each case also contains the default variable of either one or zero that represents whether the customer is expected to default or not for the next month, respectively.

C. Preprocessing

In the initial stages of data preprocessing, essential Python libraries such as NumPy and Pandas were imported to process the dataset file and perform calculations. The dataset was converted into a CSV file and read into a DataFrame object using Pandas. To gain an understanding of the features that the dataset contained, various insights about the composition of the dataset were displayed: the data types, the number of non-null values, and the memory usage of each feature. Basic statistical information was also computed such as the mean, standard deviation, minimum, and maximum to comprehend the distribution and various characteristics of the data. Columns with null values and the client ID were dropped from the dataset, as these null values could potentially interfere with the accuracy of prediction models.

Before splitting the dataset into a training and testing set, it was discovered that the dataset exhibited a significant class imbalance. There were 16,143 normal cases compared to 1,564 default cases resulting in an unbalanced ratio of 10.32. To address this issue and prevent biased model training, undersampling was used on the normal cases to output a ratio of 3:1 compared to the default cases and maintain the original asymmetry in the number of normal and default cases. Feature standardization and normalization was performed using the StandardScaler library from scikit-learn, ensuring all the features had a mean of zero and a standard deviation of one.

D. Classical Neural Net

To build the classical neural network, the Sequential/Dense model architecture in the Tensorflow/Keras Libraries was utilized. This architecture used one hidden layer with 10 nodes. This specific number was chosen from empirical observation, as there has been some research indicating that it is effective to have a hidden layer that is composed of approximately

$$nodes = \frac{input + output}{2}$$

This choice would strike a balance between model complexity and performance. To introduce non-linearity into the model, the ReLU and sigmoid activation functions were employed, with the former being used to activate the input layer as it has shown much promise in solving a variety of problems such as the vanishing gradient issue, and the sigmoid function was used to activate the outputs of the hidden layer; the sigmoid activation function was adopted as it has also shown promise in binary classification tasks. The sigmoid activation allows for the transformation of model outputs into probabilities, thereby aiding the interpretation of results in the context of binary classification. To optimize model performance, the binary cross entropy loss function was used as it was specifically designed for binary classification tasks. The Adam optimizer and accuracy metric was also utilized, being popular with this task. When it was noticed that the loss started to stagnate after the progression of epochs, the learning rate was increased. However, this led to the optimal minimum being skipped as the gradient descent algorithm would jump sporadically and output poor results. To counter this, a dynamic learning rate that exponentially decayed over time was implemented. This led to a more optimal solution; the model did not take too long to approach the minimum, and as it did approach the minimum, it started taking smaller stops so it did not overshoot. This dynamic learning rate was made using Keras' Callback LearningRateScheduler along with a custom designed exponential method.

E. Quantum Neural Net

The first section of the hybrid model architecture consisted of reducing and ordering the number of features in the original dataset using linear discriminant analysis (LDA) from 20 features to ten. LDA is used rather than principal component analysis as Mancilla and Pere [13] found that by using LDA during the data preprocessing step, quantum classifiers achieve better encoding and performance. Resultantly, the ten features retained a variability of approximately 97.67%. The output variables were also one hot encoded. These ten features were fed into a fully connected classical neural network with two hidden layers and an ELU activation function to introduce non-linearity.

The quantum layers of the architecture consist of the variational circuit. Encoding for the previous classical layer output used amplitude embedding from the Pennylane library. The variational circuit consists of the schematic of an interferometer, squeezer, another interferometer, followed by displacement and Kerr gates. The interferometers consist of rotation and beam splitters which act as matrix operations [5]. A method was also created to initialize the weights and parameters to be optimized in order to convert each quantum layer into a Keras layer for the model. The circuit was created using

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the Strawberry Fields Tensorflow device from the Pennylane library. The overall quantum schematic was repeated for five blocks and the measurement results were fed into a classical optimizer (Adam) to tune the parameterized gates in the circuit.

IV. Results and Analysis

Both models in the experiment were run for at least 30 epochs with ten percent of the overall data being used as testing data. Additionally, the QCNN was allowed to run for an additional ten epochs (40 epochs total). As both models were being trained, the accuracy and loss were outputted for both the training and the validation data in each epoch. At the 30 epoch mark, the classical neural network outputted a training accuracy of 91.22% with a loss value of 0.2627. Similarly, the validation set returned an accuracy of 91.11% with a loss value of 0.3128. The accuracy and loss plot for the classical model can be found in Fig. 1. and Fig. 2., respectively. At the

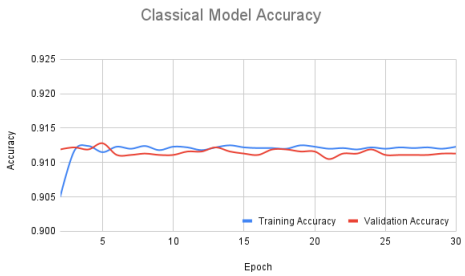

Fig. 1. Training and validation accuracy for classical model

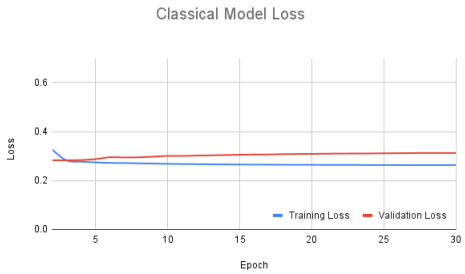

Fig. 2. Training and validation loss for classical model

same epoch point, the QCNN returned a better accuracy value of 96.34% for the training set and 95.18% for the validation set, as shown in Fig. 4. The loss values were also better with values of 0.0354 and 0.0487, respectively (exhibited in Fig. 5). With an additional ten epoch run, the QCNN was able to achieve an even higher accuracy of 97.63% for the training set and 96.21% for the validation set. The loss values were 0.0321 and 0.0451, respectively.

It was also observed that the QCNN took a significantly higher time to run per epoch with an average run time of 70 seconds per step. On the other hand, the classical neural network ran with an average run time of seven seconds per

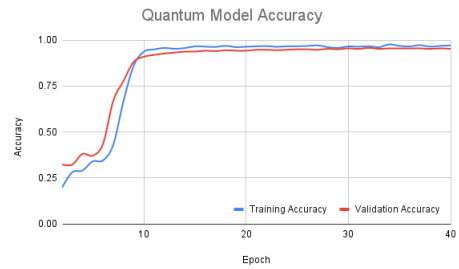

Fig. 3. Training and validation accuracy for quantum model

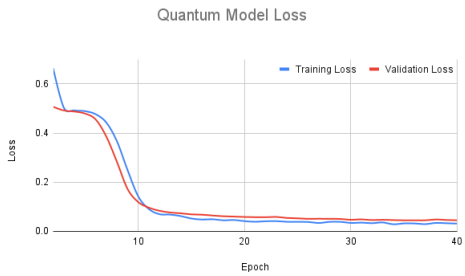

Fig. 4. Training and validation loss for quantum model

step. Moreover, it was observed that the quantum model had higher variability in both accuracy and loss compared to the classical model. For instance, at the same accuracy interval in which the classical model tended to reach its' maximum accuracy, the quantum model's accuracy was still rapidly increasing as shown in Fig. 3. Simultaneously, beginning from

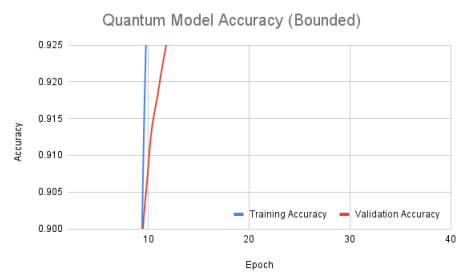

Fig. 5. Training and validation accuracy for quantum model bounded to same vertical axis values (0.9 - 0.925) as classical model

the first epoch for the classical model it's loss was constricted between the interval 0.2 and 0.4 throughout the duration of the run time. On the other hand, the quantum model's loss started to level out after the 15th epoch with high variability before then.

V. Conclusion

This paper presented a strong proof of concept for the integration of quantum technology in pre-existing models for financial classification tasks. With the hybridization of quantum technology with classical models, it allows for quantum

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technology to take advantage of quantum mechanical properties to perform complex linear transformations and classical technology to perform non-linear transformations. Subsequent investigations could explore employing an advanced feature map or ansatz in the variational quantum circuits utilized by the neural network. Furthermore, conducting a more comprehensive examination of hyperparameter optimization, specifically focusing on the learning rate and batch size, might yield beneficial outcomes in enhancing results on contemporary quantum hardware. Lastly, exploring larger and more intricate hybrid neural network architectures could be a worthwhile pursuit for tackling more demanding classification problems.

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